



BirdSound3D - AI201

Interactive Spectrographic Modeling and AI Classification of Bird Calls

Course: AI201

Instructor: Aamir Khan Maroofi

Institution:

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

Student Name: Atta Ur Rahman

Roll Number: 2024122

Project Report

Abstract

Bird vocalizations carry rich information about species identity, behavior, and ecosystem health. Manual analysis of such audio data is time-consuming and requires expert knowledge, while many existing automated systems focus solely on classification and provide limited interpretability. This project, *BirdSound3D-AI201*, presents an integrated system for preprocessing, feature extraction, interactive three-dimensional visualization, and machine learning-based classification of bird calls. Using Python and standard audio-processing libraries, the system generates spectrogram-based 3D visualizations that allow users to intuitively explore temporal and frequency patterns in bird sounds. In addition, a Random Forest classifier is trained on extracted audio features to identify bird species. A Streamlit-based web interface enables non-technical users to upload audio files, visualize bird calls, and obtain classification results.

1 Introduction

Monitoring bird populations is an important task in biodiversity conservation and ecological research. Bird calls are widely used as non-invasive indicators of species presence and environmental change. However, manual identification from audio recordings is slow, subjective, and difficult to scale. Automated approaches based on machine learning have shown promise, but many systems act as black boxes and do not provide meaningful visual insight into the structure of the audio signals.

The goal of this project is to bridge this gap by combining AI-based classification with interactive 3D spectrographic visualization. By visually representing bird calls in three dimensions and linking these representations to classification results, the system supports both analytical understanding and automated decision-making. The project emphasizes object-oriented programming and the use of scientific Python libraries such as NumPy, Pandas, and Matplotlib to ensure efficient computation, data handling, and visualization.

2 Objectives

The main objectives of the project are:

1. To preprocess bird call audio data for consistent and reliable analysis.
2. To extract meaningful audio features such as spectrograms, MFCCs, and chromagrams.
3. To design and implement interactive 3D visualizations of bird vocal signatures.
4. To develop a machine learning model for bird species classification.
5. To provide a user-friendly interface for audio upload, visualization, and classification.

3 Dataset Description

Due to limited availability of labeled data during development, the project primarily uses synthetic bird call audio generated through a custom data generator. This synthetic dataset simulates frequency and temporal patterns of real bird calls while maintaining controlled variability.

- **Number of species:** 5 (Common Blackbird, European Robin, Nightingale, Garden Warbler, Tawny Owl)
- **Audio format:** WAV
- **Audio duration:** 2.5–4.0 seconds
- **Number of samples:** Approximately 10 audio files per species

Support for loading local audio files and integration with the Xeno-canto API was implemented, allowing future expansion.

4 Methodology

4.1 Preprocessing

Audio preprocessing is a critical step to ensure consistent feature extraction and model performance. The preprocessing pipeline includes normalization, noise reduction, silence trimming, resampling to 22,050 Hz, and optional segmentation. These operations are implemented using NumPy arrays for efficient numerical computation on raw audio signals. A key challenge was reducing background noise while preserving subtle acoustic features essential for species identification.

4.2 Feature Extraction

After preprocessing, multiple audio features were extracted using the Librosa library. NumPy is used internally to represent spectrograms, MFCC matrices, and other extracted features, enabling efficient aggregation and transformation of data. The extracted features include:

- Spectrograms
- Mel-Frequency Cepstral Coefficients (MFCCs)
- Chromagrams
- Spectral contrast, zero-crossing rate, and RMS energy

Feature vectors and corresponding class labels are organized using Pandas DataFrames, which simplifies dataset management and preparation for machine learning.

4.3 3D Visualization

The extracted spectrogram data is mapped into three-dimensional space, where the x-axis represents time, the y-axis represents frequency, and the z-axis represents amplitude. Matplotlib is used to generate static 2D and 3D visualizations suitable for analysis and documentation, while Plotly is employed for interactive 3D spectrograms supporting rotation, zoom, and hover-based exploration.

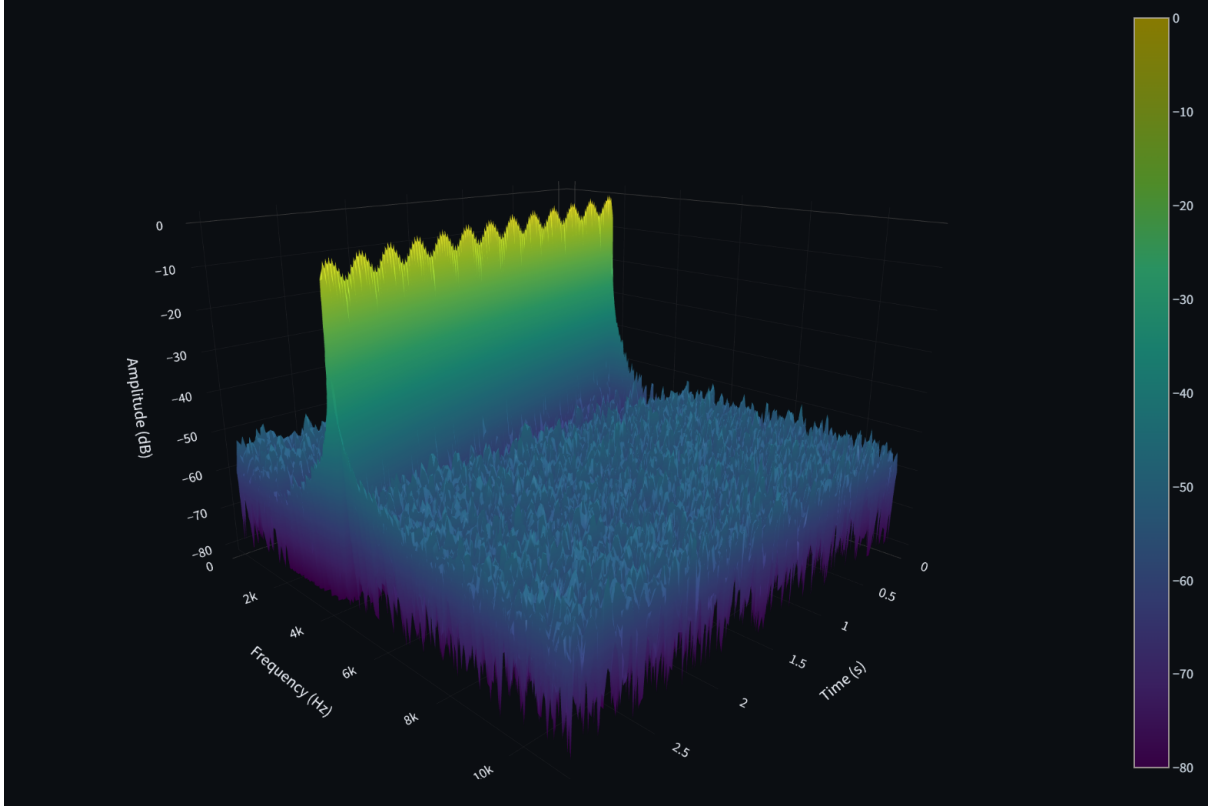


Figure 1: Interactive 3D spectrogram visualization illustrating time–frequency–amplitude relationships.

4.4 Machine Learning Classification

A Random Forest classifier was implemented using Scikit-learn and trained on aggregated feature sets prepared using Pandas. NumPy arrays are used during training and inference for efficient numerical processing. Model performance was evaluated using accuracy, precision, recall, and F1-score metrics.

5 Software Design and Implementation

The project follows an object-oriented programming approach with separate classes for data loading, preprocessing, feature extraction, visualization, and classification. Pandas and NumPy are integrated across modules for data handling and computation. A Streamlit-based web application provides multiple pages for uploading audio files, viewing 3D spectrograms, and obtaining classification results.

6 Results and Discussion

The system successfully integrates audio processing, visualization, and machine learning. Distinct frequency ranges and temporal modulations were observed across different bird species through Matplotlib and Plotly visualizations. The Random Forest classifier achieved reasonable performance on the synthetic dataset, demonstrating the feasibility of automated bird species identification.

7 Limitations

- Synthetic data may not fully capture real-world acoustic complexity.
- Classification performance is limited by the small dataset size.
- Visualization of large audio files can be computationally expensive.

8 Conclusion

BirdSound3D-AI201 presents a comprehensive system for bird call analysis using preprocessing, feature extraction, 3D visualization, and machine learning classification. The project demonstrates effective use of NumPy, Pandas, and Matplotlib alongside modern AI tools to create an interpretable and functional bioacoustic analysis system.

9 Future Work

Future enhancements include integration with large real-world datasets, deep learning models, advanced noise reduction, and real-time audio processing.

10 References

1. Librosa: Audio and Music Signal Analysis in Python.
2. Scikit-learn: Machine Learning in Python.
3. Plotly and Matplotlib Documentation.